Datasheets for Learning Traces: Documentation Practices for Educational Interaction Datasets

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Abstract

We audited the documentation of 37 publicly released learning-interaction datasets against an adapted datasheet. Fewer than a third reported how consent was obtained (30%) and fewer than a fifth stated how train and test splits were formed (19%). Only 4 reported the share of students who opted out. We propose a 12-question supplement for learning traces covering consent, granularity, de-identification, split units and known artifacts of the collection platform.

1 Introduction

General-purpose datasheets do not ask the questions that matter most for learning traces: who could see a student's data while it was collected, what granularity was logged, and whether a student can appear in both training and test data.

2 Audit

We identified 37 datasets released between 2015 and 2024 with event-level student interaction data. Two authors coded each dataset's paper and documentation.

Item	Reported
Institution and course context	92%
Consent procedure	30%
Opt-out rate	11%
De-identification method	57%
Split unit (row / student / term)	19%
Known logging gaps or outages	8%

Known logging gaps were the rarest item, though three of the dataset maintainers we contacted confirmed outages during collection.

3 Supplement

Our 12 questions include: What events were logged and at what granularity? Were there periods when logging failed, and how are they marked? What is the unit of the provided splits? Can a student appear in more than one split? What would a student need to do to have their data removed?

4 Discussion

Logging gaps deserve special attention. A reconnect bug that drops events for one lab section produces a dataset in which that section looks less active, which a model may learn as a property of the students.

References

[1] Gebru, T. et al. Datasheets for datasets. Communications of the ACM 64, 12 (2021), 86-92.